Cell-Free Massive MIMO Beamforming Based on Radio Map

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***Abstract*—In this paper, we present a framework of cell-free massive multiple-input multiple-output (MIMO) beamforming based on radio map. The radio map builds the bridge between the location of user equipment (UE) and its channel information, allowing the UE to only feed back its location to the base station (BS) / access point (AP) for beamforming, which saves significant channel state information (CSI) feedback overhead. Since the collected CSI data with position labels is limited and discrete, deep learning method is used to form the complete radio map. In addition, to compensate for the random phase gap between the coordinated BSs of cell-free MIMO, which is more significant when merely location information is reported, space-time block coding (STBC) is employed. According to the numerical simulations, the proposed radio map method is superior to location-based beamforming (LBB) and channel knowledge map (CKM) methods. The numerical results also demonstrate that carrier aggregation (CA) with multi-BS coordination empowered by STBC can improve communication performance greatly.**

***Index Terms*—Beamforming, cell-free massive MIMO, deep learning, radio map.**

I. Introduction

In this paper, we consider beamforming in cell-free massive multiple-input multiple-output (MIMO) [1], [2], where multiple access points (APs), i.e., base stations (BSs) serve multiple user equipment (UE) in the same time-frequency resource cooperatively. In the assumption of the cell-free network, all the BSs are connected to the central controller (CC) via a backhaul network to achieve collaboration. For a specific UE, being served by multiple BSs can provide extra diversity gain against fading or multiplexing gain for throughput. Besides, compared with conventional cellular networks, there are no cells or cell boundaries in cell-free massive MIMO [1], which avoids handoff and improves communication stability. Therefore, cell-free massive MIMO is considered a very promising technology in the future [3].

In general, instantaneous channel state information (CSI) is required to perform beamforming in MIMO systems. To alleviate or even eliminate the dependence of beamforming on CSI, we introduce the concept of radio map [4]. Initially, radio map is defined as the geographical signal power spectrum density as a function of location, frequency and time [4], but later contains more channel-related information like pathloss, time of arrival (ToA), phase, azimuth and zenith angle of departure (AoD) and angle of arrival (AoA) [5]–[7]. Consequently, radio map can be seen as the bridge connecting channel knowledge and the information of UE. As is known, many applications in wireless communications rely heavily on channel knowledge to achieve better performance [8], which makes radio map a critical technology.

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A. Related Works and Motivations

Many papers have studied beamforming for cell-free massive MIMO such as conjugate beamforming [1], [2] and zero-forcing [9]. However, in their assumptions, all BSs are equipped with a single antenna, which is far from the practice. In addition, CSI is indispensable for beamforming in their works. In time-division duplex (TDD) systems, CSI can be acquired by leveraging the channel reciprocity while in frequency-division duplex (FDD) systems, UE needs to provide CSI feedback to the BS, which incurs huge uplink overhead.

In order to mitigate the reliance on CSI feedback, some works introduce location information to assist beamforming [10]–[12]. The main idea of location-based beamforming (LBB) is to calculate the AoD from the BS to the UE based on their locations. Then, the transmit precoding vector can be designed with the obtained AoD to align the beam with the location of the UE. The LBB method works well under line-of-sight (LoS) conditions but deteriorates visibly in non-LoS (NLoS) situations.

To address the issue of NLoS, channel knowledge map (CKM) [5]–[7] is built to estimate the AoA of BS. The CKM can be seen as a large database that stores the potential positions of UE and the corresponding AoAs. After getting the UE's location, K nearest neighbors (KNN) and inverse distance weighting (IDW) methods are applied to do interpolation for AoA. In this method, all measurement data need to be saved to perform interpolation, which places strict demands on device memory. In addition, KNN algorithm requires distance calculation and sorting, which brings a substantial computational burden when the size of the measurement data is large.

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B. Contributions and Paper Organization

Motivated by the above discussions, this paper makes the following specific contributions.

- We present a framework of cell-free massive MIMO beamforming based on radio map. In this framework, UE only needs to feed back its location instead of CSI to the BS, which saves significant CSI feedback overhead.
- We propose a radio map method for cell-free massive MIMO beamforming based on deep learning. In the proposed method, deep neural network (DNN) is utilized to build the radio map according to the collected CSI data with position labels.
- We study the collaboration between multiple BSs, applying carrier aggregation (CA) to improve spectral efficiency (SE), and employing space-time block coding (STBC) to compensate for the random phase gap. We also derive the expressions of SE in applying CA and STBC, respectively.
- We validate the superiority of the deep learning-based radio map method compared with LBB method [10]–[12] and CKM method [5]–[7]. We also demonstrate that CA can greatly improve SE, and employing STBC can obtain further improvement by numerical simulations.

The remainder of the paper is organized as follows. In Section II, the system model of cell-free massive MIMO beamforming based on radio map is presented. In Section III, the deep learning-based radio map method for beamforming is proposed. In Section IV, the collaboration between BSs by CA and STBC is studied. In Section V, numerical simulations and results are given, followed by conclusions in Section VI.

Notations: a, $\mathbf{a}$ and $\mathbf{A}$ represent a scalar, a vector and a matrix, respectively. $(\cdot)^*$, $(\cdot)^T$ and $(\cdot)^H$ represent conjugate, transpose and conjugate transpose operation, respectively. $\lceil\cdot\rceil$ is the ceiling operation and $||\cdot||$ denotes the ℓ_2 norm. $\mathcal{CN}(\mu,\sigma^2)$ represents complex Gaussian distribution with the mean μ and the standard deviation σ. $\mathbf{0}$ is the zero vector and $\mathbf{I}$ is the identity matrix. $\mathbb{C}$ stands for the set of complex numbers.

II. System Model

Consider the scenario of cell-free massive MIMO beamforming based on radio map as shown in Fig. 1. In the downlink transmission, there are M BSs each equipped with N_t transmit antennas and all connected to the CC through the backhaul link. The hexagons in the figure represent the cells in the classical cellular network, and the dashed line here indicates that there are no cell boundaries existing in this cell-free network. Thus, UE can be served by multiple BSs instead of selecting the best of these connections. In this paper, we focus on the communication performance of a single UE within a specific area and the number of possible locations for UE is denoted by N. The UE has N_r antennas for receiving the information from BSs and can get its location from the global position system (GPS). Without loss of generality, orthogonal frequency division multiplexing (OFDM) is adopted for transmission, and the number of subcarriers is K.

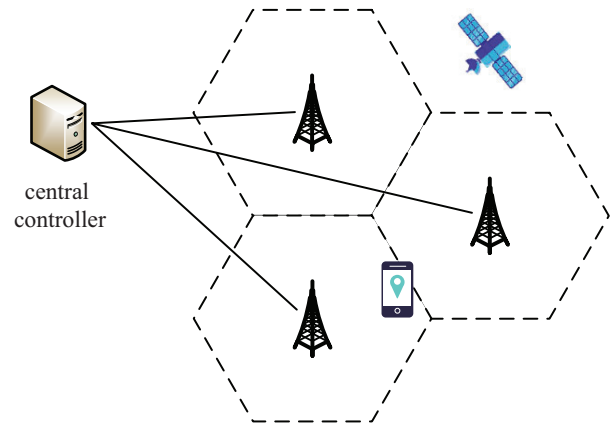

Fig. 1: The scenario of cell-free massive MIMO beamforming based on radio map.

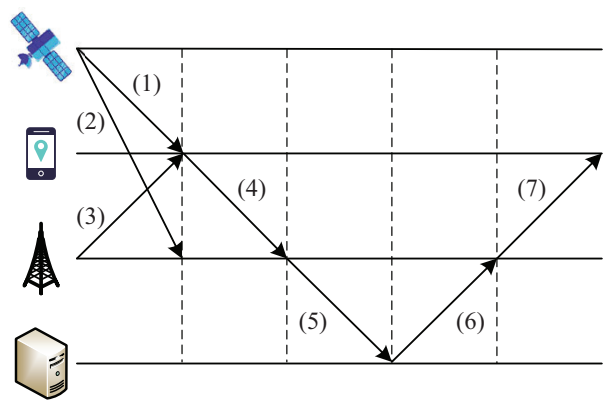

Fig. 2: The online procedure of the framework.

The framework of cell-free massive MIMO beamforming based on radio map can be divided into offline and online parts. In the offline part, deep learning is applied to build the radio map based on the measured downlink CSI with position labels. Then, the built radio map can be used in the online part to generate the transmit precoding vector of BS based on the location of UE. The online procedure is shown in Fig. 2, and the main steps are summarized as follows.

(1) The satellites send the GPS signal to the UE so that the UE can obtain its location.
(2) The satellites also send the GPS signal to the BS for correction.
(3) The BS broadcasts to the UE within its coverage.
(4) The UE responds to the BS and then sends its location.
(5) The BS calculates the more accurate differential GPS (DGPS) position of the UE and then sends to the CC.
(6) The CC allocates BSs to serve the UE according to the received location and then sends the control signals to the corresponding BSs.
(7) The assigned BSs compute their transmit precoding vectors to perform beamforming based on the built radio maps to serve the UE cooperatively.

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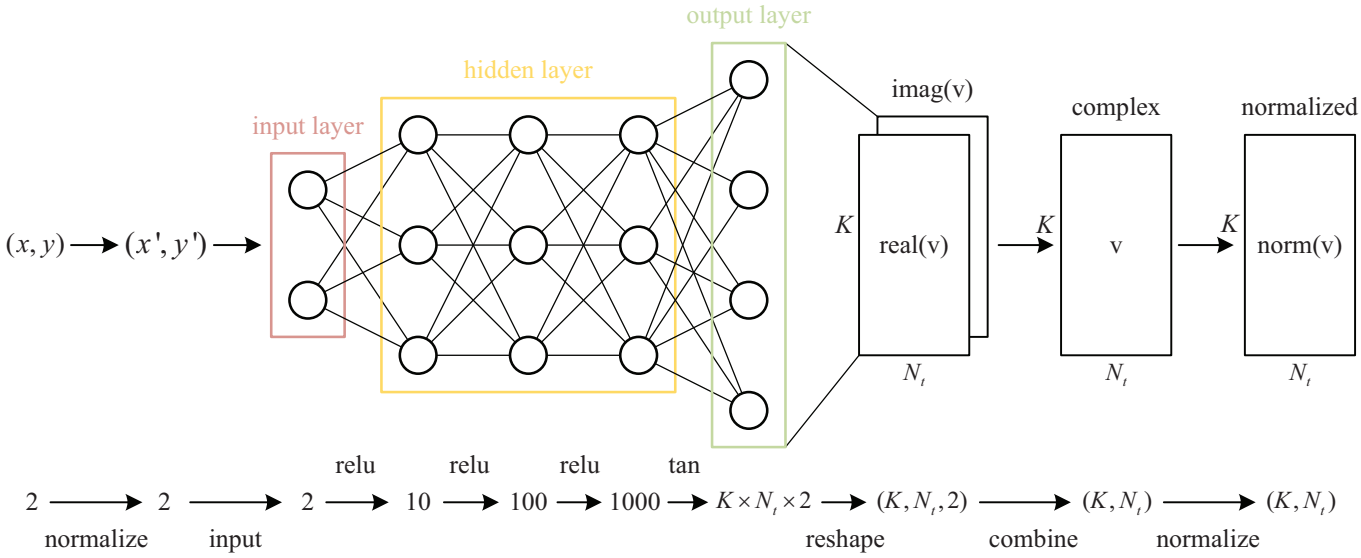

Fig. 3: The neural network structure.

III. Radio Map Method for Beamforming Based on Deep Learning

In this section, we build the radio map for beamforming based on deep learning, which can be used to output the transmit precoding vector $\mathbf{v}$ by taking the location $\mathbf{p}$ as input, i.e.,

$$\mathbf{v} = \mathcal{F}\left(\mathbf{p}; \theta\right), \tag{1}$$

where $\mathbf{p} = (x, y)$ represents the two-dimensional coordinates of UE, and θ is the set of the parameters of the neural network. DNN is applied to form a complete radio map with limited and discrete collected data.

A. Neural Network

To build the radio map, DNN is adopted which is proved to be able to approximate any function that has continuous values by universal approximation theory [13]. Meanwhile, DNN is one of the most basic neural network models in deep learning, which indicates that there is no need to deliberately develop a complex model to achieve satisfactory performance.

The neural network structure is shown in Fig. 3. As shown in the figure, the normalized location of UE with the range of $(0, 1)$ is fed to the input layer, and rectified linear unit (ReLU) [14] is employed as the activation function. Then, multiple hidden layers are utilized to expand the dimension of the input data and learn the relationship between the position and the precoding vector. Next, the output layer applies hyperbolic tangent (tanh) activation function to scale the output to the $(-1, 1)$ range. After that, the result of the output layer can be reshaped to obtain the real and imaginary parts of the precoding vector, which can be combined to form the complex precoding vector. Finally, normalization is performed to meet the power constraint.

B. Customized Loss Function

To improve the performance of the DNN, we customized the loss function. In centralized training, each BS needs to send the collected CSI data with position labels to the CC, which requires a tremendous amount of communication overhead. Therefore, distributed training is considered here, where every BS wishes to maximize its downlink SE to UE, so the loss function can be customized as [15]

$$\begin{aligned} &\mathcal{L}\Big(\mathcal{F}\left(\mathbf{p}_n; \theta_m\right), \mathbf{H}_n^m\Big) \\ &= -\sum_{k=1}^{K} \frac{1}{K} \log\left(1 + \frac{\left|\left|\mathbf{H}_{nk}^m \mathcal{F}_k\left(\mathbf{p}_n; \theta_m\right)\right|\right|^2}{\sigma^2}\right), \end{aligned} \tag{2}$$

where $\mathbf{p}_n$ is the coordinates of the nth location of UE, θ_m denotes the set of the parameters of the neural network at the mth BS, and $\mathbf{H}_n^m$ is the CSI from the mth BS to the UE at the nth location. The training objective is to minimize the expected loss function of the UE at all potential locations, i.e.,

$$\theta_m^* = \arg\min_{\theta_m} \frac{1}{N} \sum_{n=1}^{N} \mathcal{L}\Big(\mathcal{F}(\mathbf{p}_n; \theta_m), \mathbf{H}_n^m\Big). \tag{3}$$

It has been proved in [15] that the customized loss function in (2) is differentiable with respect to θ_m, which provides feasibility for gradient calculation and weight update.

IV. The Collaboration between BSs

In this section, we study the collaboration between multiple BSs by applying CA. To compensate for the random phase gap in CA, STBC is employed to address this issue. We derive the expressions of SE under the circumstances of applying CA and STBC compared with frequency division multiplexing access (FDMA).

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A. FDMA

In FDMA [16], M BSs jointly serve the UE by allocating different frequency band resources. The received signal of the UE on the kth subcarrier can be written as

$$\mathbf{y}_k^{\text{FDMA}} = \mathbf{H}_k^m \mathbf{x}_k^m + \mathbf{n} = \mathbf{H}_k^m \mathbf{v}_k^m s_k + \mathbf{n}, \tag{4}$$

where $\mathbf{y}_k^{\text{FDMA}} \in \mathbb{C}^{N_r \times 1}$ represents the received signal of the kth subcarrier, $\mathbf{H}_k^m \in \mathbb{C}^{N_r \times N_t}$ is the downlink CSI transmitted from the mth BS and $m = \left\lceil \frac{kM}{K} \right\rceil$. $\mathbf{x}_k^m = \mathbf{v}_k^m s_k \in \mathbb{C}^{N_t \times 1}$ is the transmitted signal where $\mathbf{v}_k^m \in \mathbb{C}^{N_t \times 1}$ is the transmit precoding vector and s_k is a complex number which denotes the symbol for transmission, $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_{N_r}) \in \mathbb{C}^{N_r \times 1}$ is the complex additive Gaussian noise.

The SE of the kth subcarrier can be computed as

$$\mathbf{R}_k^{\text{FDMA}} = \log\left(1 + \frac{\left|\left|\mathbf{H}_k^m \mathbf{v}_k^m\right|\right|^2}{\sigma^2}\right) \tag{5}$$

and the average SE of the UE at all locations over all subcarriers is given by

$$\begin{aligned}\overline{\mathbf{R}}^{\text{FDMA}} &= \frac{1}{NK} \sum_{n=1}^{N} \sum_{m=1}^{M} \sum_{k=k'}^{k''} \mathbf{R}_k^{\text{FDMA}} \\ &= \frac{1}{NK} \sum_{n=1}^{N} \sum_{m=1}^{M} \sum_{k=k'}^{k''} \log\left(1 + \frac{\left|\left|\mathbf{H}_{nk}^m \mathbf{v}_{nk}^m\right|\right|^2}{\sigma^2}\right),\end{aligned} \tag{6}$$

where $k' = \frac{(m-1)K}{M} + 1$ and $k'' = \frac{mK}{M}$ denote the first and the last subcarrier assigned to the mth BS, respectively.

B. The Collaboration by CA

To fully utilize the spectrum resources, CA [17], [18] is introduced here to achieve the collaboration between BSs. Through CA, multiple BSs can share the spectrum resources that were previously allocated to a single BS, which enhances spectral utilization efficiency significantly. In this way, multiple BSs can transmit to the UE simultaneously on each subcarrier. To avoid interference, BSs need to cooperate to transmit identical symbols on the same subcarrier. The received signal at the UE on the kth subcarrier from M BSs can be expressed as

$$\mathbf{y}_k^{\text{CA}} = \sum_{m=1}^{M} \mathbf{H}_k^m \mathbf{x}_k^m + \mathbf{n} = \sum_{m=1}^{M} \mathbf{H}_k^m \mathbf{v}_k^m s_k + \mathbf{n}, \tag{7}$$

The spectral efficiency (SE) of the kth subcarrier can be computed as

$$\mathbf{R}_k^{\text{CA}} = \log\left(1 + \frac{\left|\left|\sum_{m=1}^{M} \mathbf{H}_k^m \mathbf{v}_k^m\right|\right|^2}{\sigma^2}\right) \tag{8}$$

and the average SE of the UE at all locations over all subcarriers is

$$\begin{aligned}\overline{\mathbf{R}}^{\text{CA}} &= \frac{1}{NK} \sum_{n=1}^{N} \sum_{k=1}^{K} \mathbf{R}_k^{\text{CA}} \\ &= \frac{1}{NK} \sum_{n=1}^{N} \sum_{k=1}^{K} \log\left(1 + \frac{\left|\left|\sum_{m=1}^{M} \mathbf{H}_{nk}^m \mathbf{v}_{nk}^m\right|\right|^2}{\sigma^2}\right).\end{aligned} \tag{9}$$

Remark 1: Notice that in (9) the signals from different BSs are superimposed, which requires compensation for random phase gap to avoid signal cancellation.

C. Compensating for random phase gap by STBC

To compensate for random phase gap, STBC is employed. For simplicity, two BSs are considered and Alamouti code [19] is applied here which has the form of

$$\mathbf{C} = \begin{bmatrix} s_1 & s_2 \\ -s_2^* & s_1^* \end{bmatrix}. \tag{10}$$

Then, the received signal of the UE on the kth subcarrier at the first time slot is

$$\mathbf{y}_{k1}^{\text{STBC}} = \mathbf{H}_k^1 \mathbf{v}_k^1 s_1 + \mathbf{H}_k^2 \mathbf{v}_k^2 s_2 + \mathbf{n}_1 \tag{11}$$

while the second time slot is

$$\mathbf{y}_{k2}^{\text{STBC}} = \mathbf{H}_k^1 \mathbf{v}_k^1 (-s_2^*) + \mathbf{H}_k^2 \mathbf{v}_k^2 s_1^* + \mathbf{n}_2. \tag{12}$$

(11) and (12) can be combined in matrix form as

$$\mathbf{y}_k^{\text{STBC}} = \begin{bmatrix} \mathbf{y}_{k1}^{\text{STBC}} \\ \mathbf{y}_{k2}^{\text{STBC}} \end{bmatrix} = \begin{bmatrix} s_1 & s_2 \\ -s_2^* & s_1^* \end{bmatrix} \begin{bmatrix} \mathbf{H}_k^1 \mathbf{v}_k^1 \\ \mathbf{H}_k^2 \mathbf{v}_k^2 \end{bmatrix} + \begin{bmatrix} \mathbf{n}_1 \\ \mathbf{n}_2 \end{bmatrix}. \tag{13}$$

Take negative complex conjugate of the second received sample as

$$\begin{aligned}\tilde{\mathbf{y}}_k^{\text{STBC}} &= \begin{bmatrix} \mathbf{y}_{k1}^{\text{STBC}} \\ -\mathbf{y}_{k2}^{\text{STBC}*} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{H}_k^1 \mathbf{v}_k^1 & \mathbf{H}_k^2 \mathbf{v}_k^2 \\ -\left(\mathbf{H}_k^2 \mathbf{v}_k^2\right)^* & \left(\mathbf{H}_k^1 \mathbf{v}_k^1\right)^* \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} + \begin{bmatrix} \mathbf{n}_1 \\ -\mathbf{n}_2^* \end{bmatrix} \\ &= \mathbf{H}_A \mathbf{s} + \tilde{\mathbf{n}},\end{aligned} \tag{14}$$

where $\mathbf{s} = [s_1, s_2]^T$, $\tilde{\mathbf{n}} = [\mathbf{n}_1, -\mathbf{n}_2^*]^T$ and

$$\mathbf{H}_A = \begin{bmatrix} \mathbf{H}_k^1 \mathbf{v}_k^1 & \mathbf{H}_k^2 \mathbf{v}_k^2 \\ -\left(\mathbf{H}_k^2 \mathbf{v}_k^2\right)^* & \left(\mathbf{H}_k^1 \mathbf{v}_k^1\right)^* \end{bmatrix}. \tag{15}$$

It can be seen from (15) that $\mathbf{H}_A$ satisfies

$$\mathbf{H}_A^H \mathbf{H}_A = \left(\left|\left|\mathbf{H}_k^1 \mathbf{v}_k^1\right|\right|^2 + \left|\left|\mathbf{H}_k^2 \mathbf{v}_k^2\right|\right|^2\right) \cdot \mathbf{I}_{2N_r} = a \cdot \mathbf{I}_{2N_r}. \tag{16}$$

The decoding can be performed by left-multiplying $\mathbf{H}_A^H$ to (14) as

$$\mathbf{H}_A^H \tilde{\mathbf{y}}_k^{\text{STBC}} = \mathbf{H}_A^H \mathbf{H}_A \mathbf{s} + \mathbf{H}_A^H \tilde{\mathbf{n}} = a\mathbf{s} + \mathbf{w}, \tag{17}$$

where $\mathbf{w} = \mathbf{H}_A^H \tilde{\mathbf{n}} \sim \mathcal{CN}(\mathbf{0}, a\sigma^2 \mathbf{I}_{2N_r})$.

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According to (17), the SE of the kth subcarrier can be written as

$$\mathbf{R}_k^{\text{STBC}} = \log\left(1 + \frac{\left|\left|\mathbf{H}_k^1\mathbf{v}_k^1\right|\right|^2 + \left|\left|\mathbf{H}_k^2\mathbf{v}_k^2\right|\right|^2}{\sigma^2}\right). \tag{18}$$

The two BSs in (18) can be extended to M BSs as

$$\mathbf{R}_k^{\text{STBC}} = \log\left(1 + \frac{\sum_{m=1}^{M}\left|\left|\mathbf{H}_k^m\mathbf{v}_k^m\right|\right|^2}{\sigma^2}\right) \tag{19}$$

and the average SE of the UE at all locations over all subcarriers is calculated as

$$\begin{aligned}\overline{\mathbf{R}}^{\text{STBC}} &= \frac{1}{NK}\sum_{n=1}^{N}\sum_{k=1}^{K}\mathbf{R}_k^{\text{STBC}} \\ &= \frac{1}{NK}\sum_{n=1}^{N}\sum_{k=1}^{K}\log\left(1 + \frac{\sum_{m=1}^{M}\left|\left|\mathbf{H}_{nk}^m\mathbf{v}_{nk}^m\right|\right|^2}{\sigma^2}\right).\end{aligned} \tag{20}$$

Remark 2: It can be seen from (20) that the random phase gap issue in (9) can be compensated by employing STBC.

V. Numerical Simulations and Results

In this section, we present the chosen dataset and give the detailed parameters for numerical simulations. Then, simulation results are illustrated and discussions are provided.

A. Dataset and Parameters

In the numerical simulations, the DeepMIMO dataset [20] generated by Remcom Wireless InSite [21] is adopted. To make the scenario more representative, the 'O1 Blockage' scenario is chosen, which is shown in Fig. 4. In this scenario, three surfaces are added to make the channel environment more complex. To be specific, the red line in the figure is the surface to block BS and UE, while two orange lines are the other surfaces acting as reflectors to enable more NLoS paths. The area we select is represented by the red line rectangle in Fig. 4. Two BSs with indices 1 and 3 are chosen so that $M = 2$. The range of the locations of UE is from row 300 to row 1299 and the row sampling is 0.1, totalling 100 rows. There are 181 locations in each row, and we randomly sample 20 sites per row. Therefore, a total of 2000 locations are sampled, i.e., $N = 2000$.

The communication parameters are set as follows. The frequency f_c is 3.5GHz and the bandwidth B is 100MHz. The number of subcarriers K is set to 64. The transmit and receive antennas are both uniform linear array (ULA) with $N_t = 32$ and $N_r = 2$. The noise power σ^2 of 100MHz bandwidth is assumed to be -94dBm, approximately 3.9811×10^{-10}W. Besides, we define the ratio of the number of training set samples to the total number of samples as training ratio α.

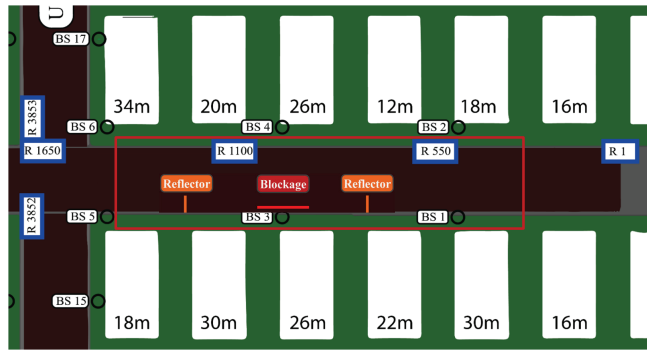

Fig. 4: The top view of the 'O1 Blockage' scenario [20].

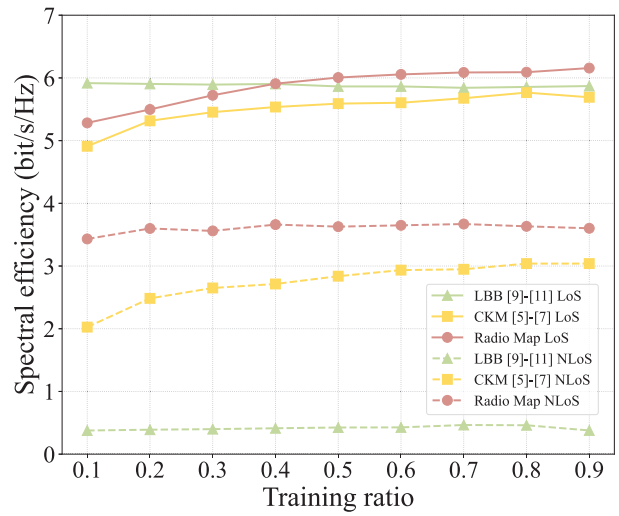

Fig. 5: The SEs of LBB, CKM and radio map methods under LoS and NLoS conditions versus training ratio α.

As for the training, the network is optimized by the Adam optimizer [22] with a learning rate of 1×10^{-4}. The number of training epochs is set to 2000, and the batch size is 128.

B. Numerical Results

To validate the proposed deep learning-based radio map method, we apply LBB method [10]–[12] and CKM method [5]–[7] as benchmarks and take both LoS and NLoS conditions into consideration. The SEs of LBB, CKM and radio map methods under LoS and NLoS conditions versus training ratio α are shown in Fig. 5, where CA with STBC is applied to achieve collaboration between two BSs. The performances of CKM method and the radio map method are related to α while LBB method is independent of α. It can be seen from the figure that under LoS condition, the SEs of CKM method and the radio map method both increase with α while the SE of LBB method remains almost unchanged. In particular, when α is larger than 0.4, the SE of the radio map method exceeds that of LBB method. In the case of NLoS, the LBB method is almost ineffective due to obstruction. The CKM method estimates the angle through the stored AoA information in

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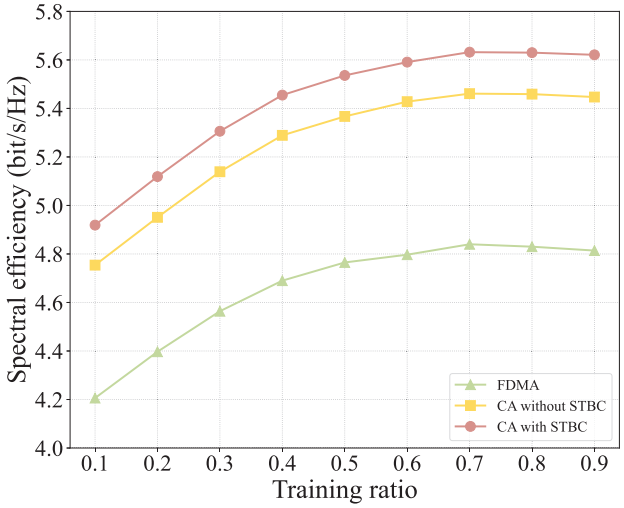

Fig. 6: The SEs of applying FDMA, CA without STBC and CA with STBC of radio map method versus training ratio α.

the database. As α grows, the estimated angle becomes more accurate, and SE increases accordingly. In NLoS situations, the radio map method is obviously better than these two methods.

In Fig. 6, we show the SE of FDMA, as well as the SEs of cooperatively serving UE by CA without STBC and CA with STBC of the radio map method versus training ratio α. In general, the three SEs in Fig. 6 all increase with α, which indicates the effectiveness of the deep learning-based radio map method under different circumstances. When $\alpha > 0.6$, the SEs of the radio map method change little with α. In this case, the continuous increase of training samples does not bring more improvements to the communication performance. Viewed separately, FDMA has the worst performance among them due to insufficient utilization of spectrum resources. After applying CA to achieve the collaboration of BSs, the SE can be improved greatly. Moreover, employing STBC can obtain a further increment of about 0.2 bit/s/Hz compared to CA without STBC at the same training ratio.

VI. Conclusion

In this paper, we first present a framework of cell-free massive MIMO beamforming based on radio map. In the framework, UE only needs to feed back its location instead of CSI to the BS, which saves significant CSI feedback overhead. Then, the radio map method for cell-free massive MIMO beamforming based on deep learning is proposed. After that, we study the collaboration between BSs and apply CA to fully exploit spectrum resources. Moreover, to address the random phase gap issue introduced by CA, we propose to employ STBC to compensate it. Simulation results indicate the superiority of the radio map method under both LoS and NLoS conditions compared with LBB and CKM methods. The numerical results also reveal that CA can improve SE greatly compared with FDMA, and employing STBC can obtain further improvement.

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